

# Survival

Frederick De Baene

## Survival

The survival function  $S(t)$  calculates the probability that the event occurs after time  $t$ :

$$S(t) = \mathbb{P}[T > t]$$

Given a sample dataset, we can estimate the survival function  $S(t)$  with the following estimator:

$$\tilde{S}(t) = \frac{\sum I[T > t]}{n}$$

with  $I[T > t]$  an indicator variable which is 1 if the event occurs after time  $t$  (and 0 otherwise),  $n$  the sample size, and  $\sum I[T > t]$  the number of observations in the dataset for which the event occurs after time  $t$ .

In this sense, the estimator represents the fraction of patients that is event-free at time  $t$ .

## No Censoring

```
# Data from the Ohio State University Bone Marrow Transplant Unit
dat <- tibble(
  date = c(1, 2, 2, 2, 3, 5, 6, 7, 8, 16, 17, 34),
  cens = c(1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1)
)
```

| date | cens |
|------|------|
| 1    | 1    |

| date | cens |
|------|------|
| 2    | 1    |
| 2    | 1    |
| 2    | 1    |
| 3    | 1    |
| 5    | 1    |
| 6    | 1    |
| 7    | 1    |
| 8    | 1    |
| 16   | 1    |
| 17   | 1    |
| 34   | 1    |

For the dataset, we can estimate  $S(t)$  using  $\tilde{S}(t)$ :

```
# Use a naive non-parametric estimator to estimate the survival function
observation_months <- 0L:34L
S_tilde <- rep(NA, max(observation_months))

for (t in observation_months) {
  S_tilde[t + 1] <- cumprod(sum(dat$date > t) / nrow(dat))
}

km_estimates <- tibble(t = observation_months, S_tilde = S_tilde)
```

To estimate the true survival function  $S(t)$ , there is also the non-parametric Kaplan-Meier (KM) estimator:

$$\hat{S}(t) = \prod_{j; t_j < t} \left(1 - \frac{d_j}{r_j}\right)$$

with  $r_j$  the number at risk at time  $t_j$ ,  $d_j$  the number of events that occur at time  $t_j$ .

```
# For each time interval (month), get the number at risk, the number of events,
# and the number of censored observations
r_j <- rep(NA, max(observation_months))
d_j <- rep(NA, max(observation_months))
c_j <- rep(NA, max(observation_months))

for (t in observation_months) {
```

```

r_j[t + 1] <- sum(dat$date >= t)
d_j[t + 1] <- sum(dat$date == t & dat$cens == 1)
c_j[t + 1] <- sum(dat$date == t & dat$cens == 0)
}

km_estimates <- add_column(
  km_estimates,
  r_j = r_j,
  d_j = d_j,
  c_j = c_j,
  .before = "S_tilde"
)

# Calculate the hazard and estimate S(t)
km_estimates$hazard <- km_estimates$d_j / km_estimates$r_j
km_estimates$S_t <- cumprod(1 - km_estimates$hazard)

```

If there is no censoring, then  $\tilde{S}(t)$  and  $\hat{S}(t)$  are the same.

| t  | r_j | d_j | c_j | S_tilde   | hazard    | S_t       |
|----|-----|-----|-----|-----------|-----------|-----------|
| 0  | 12  | 0   | 0   | 1.0000000 | 0.0000000 | 1.0000000 |
| 1  | 12  | 1   | 0   | 0.9166667 | 0.0833333 | 0.9166667 |
| 2  | 11  | 3   | 0   | 0.6666667 | 0.2727273 | 0.6666667 |
| 3  | 8   | 1   | 0   | 0.5833333 | 0.1250000 | 0.5833333 |
| 4  | 7   | 0   | 0   | 0.5833333 | 0.0000000 | 0.5833333 |
| 5  | 7   | 1   | 0   | 0.5000000 | 0.1428571 | 0.5000000 |
| 6  | 6   | 1   | 0   | 0.4166667 | 0.1666667 | 0.4166667 |
| 7  | 5   | 1   | 0   | 0.3333333 | 0.2000000 | 0.3333333 |
| 8  | 4   | 1   | 0   | 0.2500000 | 0.2500000 | 0.2500000 |
| 9  | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.2500000 |
| 10 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.2500000 |
| 11 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.2500000 |
| 12 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.2500000 |
| 13 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.2500000 |
| 14 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.2500000 |
| 15 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.2500000 |
| 16 | 3   | 1   | 0   | 0.1666667 | 0.3333333 | 0.1666667 |
| 17 | 2   | 1   | 0   | 0.0833333 | 0.5000000 | 0.0833333 |
| 18 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 19 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |

| t  | r_j | d_j | c_j | S_tilde   | hazard    | S_t       |
|----|-----|-----|-----|-----------|-----------|-----------|
| 20 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 21 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 22 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 23 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 24 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 25 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 26 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 27 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 28 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 29 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 30 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 31 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 32 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 33 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.0833333 |
| 34 | 1   | 1   | 0   | 0.0000000 | 1.0000000 | 0.0000000 |

## With Censoring

If there is censoring, then the estimator  $\tilde{S}(t)$  underestimates the probability of survival. People who are censored at time  $t$  do not die at time  $t$ . However,  $\tilde{S}(t)$  considers them as having had the event at time  $t$ .

```
# Data from the Ohio State University Bone Marrow Transplant Unit
dat <- tibble(
  date = c(1, 2, 2, 2, 3, 5, 6, 7, 8, 16, 17, 34),
  cens = c(1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 0)
)
```

The following patients were followed up for 34 months.

| date | cens |
|------|------|
| 1    | 1    |
| 2    | 1    |
| 2    | 1    |
| 2    | 0    |
| 3    | 1    |
| 5    | 1    |
| 6    | 1    |
| 7    | 0    |

| date | cens |
|------|------|
| 8    | 1    |
| 16   | 0    |
| 17   | 1    |
| 34   | 0    |

```
# Use a naive non-parametric estimator to estimate the survival function
observation_months <- 0L:34L
S_tilde <- rep(NA, max(observation_months))

for (t in observation_months) {
  S_tilde[t + 1] <- cumprod(sum(dat$date > t) / nrow(dat))
}

km_estimates <- tibble(t = observation_months, S_tilde = S_tilde)

# For each time interval (month), get the number at risk, the number of events,
# and the number of censored observations
r_j <- rep(NA, max(observation_months))
d_j <- rep(NA, max(observation_months))
c_j <- rep(NA, max(observation_months))

for (t in observation_months) {

  r_j[t + 1] <- sum(dat$date >= t)
  d_j[t + 1] <- sum(dat$date == t & dat$cens == 1)
  c_j[t + 1] <- sum(dat$date == t & dat$cens == 0)

}

km_estimates <- add_column(
  km_estimates,
  r_j = r_j,
  d_j = d_j,
  c_j = c_j,
  .before = "S_tilde"
)

# Calculate the hazard and estimate S(t)
km_estimates$hazard <- km_estimates$d_j / km_estimates$r_j
km_estimates$S_t <- cumprod(1 - km_estimates$hazard)
```

## Hazard

The hazard is the instantaneous risk of the event occurring. Heuristically, we can interpret this as a conditional probability of the event occurring at time  $t$  given that it has not yet occurred up to time  $t$ :

$$\lambda_t = \mathbb{P}[T = t | T \geq t] = \frac{d_t}{r_t}$$

with  $r_t$  the number at risk at time  $t$  and  $d_t$  the number of events that occurred at time  $t$ .

```
# Calculate the hazard
km_estimates$hazard <- km_estimates$d_j / km_estimates$r_j
```

We can now estimate the survival function with the non-parametric Kaplan-Meier estimator.

```
# Estimate S(t)
km_estimates$S_t <- cumprod(1 - km_estimates$hazard)
```

We can also estimate it using standard R packages.

```
# Estimate S(t) using the non-parametric KM estimator
km_fit <- survfit(Surv(date, cens) ~ 1, data = dat)
```

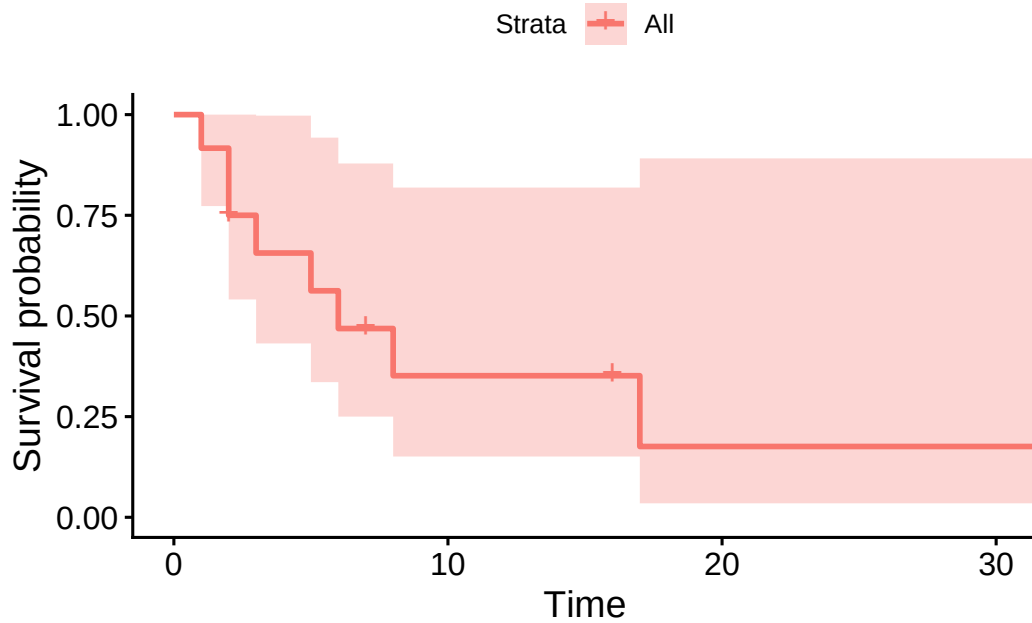
```
summary(km_fit)
```

Call: survfit(formula = Surv(date, cens) ~ 1, data = dat)

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 1    | 12     | 1       | 0.917    | 0.0798  | 0.7729       | 1.000        |
| 2    | 11     | 2       | 0.750    | 0.1250  | 0.5410       | 1.000        |
| 3    | 8      | 1       | 0.656    | 0.1402  | 0.4318       | 0.997        |
| 5    | 7      | 1       | 0.562    | 0.1482  | 0.3356       | 0.943        |
| 6    | 6      | 1       | 0.469    | 0.1503  | 0.2501       | 0.879        |
| 8    | 4      | 1       | 0.352    | 0.1517  | 0.1509       | 0.819        |
| 17   | 2      | 1       | 0.176    | 0.1456  | 0.0347       | 0.891        |

We can plot the estimated survival curve.

```
ggsurvplot(km_fit)
```



If we ignore censoring, then we underestimate the survival probability. Individuals who are censored at time  $t$  are now considered to have the event at time  $t$ .

```
# Faulty estimation of overall survival
S_tilde_t <- rep(NA, max(observation_months))

for (t in observation_months) {
  S_tilde_t[t + 1] <- cumprod(sum(dat$date > t) / nrow(dat))
}

km_estimates$S_tilde_t <- S_tilde_t
```

| t | r_j | d_j | c_j | S_tilde   | hazard    | S_t       | S_tilde_t |
|---|-----|-----|-----|-----------|-----------|-----------|-----------|
| 0 | 12  | 0   | 0   | 1.0000000 | 0.0000000 | 1.0000000 | 1.0000000 |
| 1 | 12  | 1   | 0   | 0.9166667 | 0.0833333 | 0.9166667 | 0.9166667 |
| 2 | 11  | 2   | 1   | 0.6666667 | 0.1818182 | 0.7500000 | 0.6666667 |
| 3 | 8   | 1   | 0   | 0.5833333 | 0.1250000 | 0.6562500 | 0.5833333 |
| 4 | 7   | 0   | 0   | 0.5833333 | 0.0000000 | 0.6562500 | 0.5833333 |
| 5 | 7   | 1   | 0   | 0.5000000 | 0.1428571 | 0.5625000 | 0.5000000 |
| 6 | 6   | 1   | 0   | 0.4166667 | 0.1666667 | 0.4687500 | 0.4166667 |
| 7 | 5   | 0   | 1   | 0.3333333 | 0.0000000 | 0.4687500 | 0.3333333 |

| t  | r_j | d_j | c_j | S_tilde   | hazard    | S_t       | S_tilde_t |
|----|-----|-----|-----|-----------|-----------|-----------|-----------|
| 8  | 4   | 1   | 0   | 0.2500000 | 0.2500000 | 0.3515625 | 0.2500000 |
| 9  | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.3515625 | 0.2500000 |
| 10 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.3515625 | 0.2500000 |
| 11 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.3515625 | 0.2500000 |
| 12 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.3515625 | 0.2500000 |
| 13 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.3515625 | 0.2500000 |
| 14 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.3515625 | 0.2500000 |
| 15 | 3   | 0   | 0   | 0.2500000 | 0.0000000 | 0.3515625 | 0.2500000 |
| 16 | 3   | 0   | 1   | 0.1666667 | 0.0000000 | 0.3515625 | 0.1666667 |
| 17 | 2   | 1   | 0   | 0.0833333 | 0.5000000 | 0.1757812 | 0.0833333 |
| 18 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 19 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 20 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 21 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 22 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 23 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 24 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 25 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 26 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 27 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 28 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 29 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 30 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 31 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 32 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 33 | 1   | 0   | 0   | 0.0833333 | 0.0000000 | 0.1757812 | 0.0833333 |
| 34 | 1   | 0   | 1   | 0.0000000 | 0.0000000 | 0.1757812 | 0.0000000 |

## Standard Error

The Greenwood formula can be used to estimate the standard error of survival probability estimates:

$$\hat{\sigma}^2[\hat{S}(t)] = [\hat{S}(t)]^2 \cdot \sum_{t_i \leq t} \frac{d_i}{r_i(r_i - d_i)}$$

We can calculate this manually.



$$SE[\hat{L}(t)] = \frac{SE[\hat{S}(t)]}{\hat{S}(t) \cdot |\log(\hat{S}(t))|}$$

```
# Calculate the estimated standard error using the Greenwood formula
fac_1 <- km_estimates$S_t^2
fac_2 <- cumsum(
  km_estimates$d_j
  / (km_estimates$r_j * (km_estimates$r_j - km_estimates$d_j))
)

km_estimates <- km_estimates |>
  add_column(est_var = fac_1 * fac_2) |>
  mutate(
    est_se = sqrt(est_var),
    conf_lower = S_t - 1.96*est_se,
    conf_upper = S_t + 1.96*est_se,
    adj_conf_lower = pmax(0, conf_lower),
    adj_conf_upper = pmin(1, conf_upper)
  )
```

```
# Estimate S(t) using the non-parametric KM estimator
km_fit <- survfit(
  Surv(date, cens) ~ 1,
  data = dat,
  conf.type = "plain",
  conf.lower = "usual"
)
```

```
summary(km_fit)
```

Call: `survfit(formula = Surv(date, cens) ~ 1, data = dat, conf.type = "plain", conf.lower = "usual")`

| time | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
|------|--------|---------|----------|---------|-------|--------|-------|--------|
| 1    | 12     | 1       | 0.917    | 0.0798  |       | 0.7603 |       | 1.000  |
| 2    | 11     | 2       | 0.750    | 0.1250  |       | 0.5050 |       | 0.995  |
| 3    | 8      | 1       | 0.656    | 0.1402  |       | 0.3815 |       | 0.931  |
| 5    | 7      | 1       | 0.562    | 0.1482  |       | 0.2720 |       | 0.853  |
| 6    | 6      | 1       | 0.469    | 0.1503  |       | 0.1742 |       | 0.763  |
| 8    | 4      | 1       | 0.352    | 0.1517  |       | 0.0543 |       | 0.649  |
| 17   | 2      | 1       | 0.176    | 0.1456  |       | 0.0000 |       | 0.461  |

Using this approach, we can yield lower confidence limits smaller than 0 or upper confidence limits greater than 1. We transform  $\hat{S}(t)$ :

$$\hat{L}(t) = \log(-\log(S(t)))$$

We estimate the standard error of  $\hat{L}(t)$  as follows:

```
L_t <- log(-log(km_estimates$S_t))
L_t_se <- km_estimates$est_se / (km_estimates$S_t * abs(log(km_estimates$S_t)))
L_t_lower <- L_t - 1.96*L_t_se
L_t_upper <- L_t + 1.96*L_t_se

S_t_lower <- exp(-exp(L_t_upper))
S_t_upper <- exp(-exp(L_t_lower))

S_t_lower
```

```
[1]      NaN 0.53896518 0.40840828 0.32040498 0.32040498 0.24367531
[7] 0.17615073 0.17615073 0.09563541 0.09563541 0.09563541 0.09563541
[13] 0.09563541 0.09563541 0.09563541 0.09563541 0.09563541 0.01199509
[19] 0.01199509 0.01199509 0.01199509 0.01199509 0.01199509 0.01199509
[25] 0.01199509 0.01199509 0.01199509 0.01199509 0.01199509 0.01199509
[31] 0.01199509 0.01199509 0.01199509 0.01199509 0.01199509
```

```
# Estimate S(t) using the non-parametric KM estimator
km_fit <- survfit(
  Surv(date, cens) ~ 1,
  data = dat,
  conf.type = "log-log"
)
```

```
summary(km_fit)
```

Call: `survfit(formula = Surv(date, cens) ~ 1, data = dat, conf.type = "log-log")`

| time | n.risk | n.event | survival | std.err | lower | 95% CI | upper | 95% CI |
|------|--------|---------|----------|---------|-------|--------|-------|--------|
| 1    | 12     | 1       | 0.917    | 0.0798  |       | 0.5390 |       | 0.988  |
| 2    | 11     | 2       | 0.750    | 0.1250  |       | 0.4084 |       | 0.912  |
| 3    | 8      | 1       | 0.656    | 0.1402  |       | 0.3204 |       | 0.856  |

|    |   |   |       |        |        |       |
|----|---|---|-------|--------|--------|-------|
| 5  | 7 | 1 | 0.562 | 0.1482 | 0.2437 | 0.791 |
| 6  | 6 | 1 | 0.469 | 0.1503 | 0.1762 | 0.718 |
| 8  | 4 | 1 | 0.352 | 0.1517 | 0.0956 | 0.628 |
| 17 | 2 | 1 | 0.176 | 0.1456 | 0.0120 | 0.505 |

## Comparing Curves

```
# Get a sample dataset
# Create a data frame
dat <- data.frame(
  time = c(12, 18, 24, 30, 36, 6, 15, 20, 28, 40, 8, 10, 14, 16, 22, 5, 12, 18, 20, 25),
  status = c(1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1, 0),
  group = factor(rep(c("Treatment", "Control"), each = 10))
)
```

```
km_fit <- survfit(Surv(time, status) ~ group, data = dat)
```

```
summary(km_fit)
```

Call: survfit(formula = Surv(time, status) ~ group, data = dat)

| group=Control |        |         |          |         |              |              |
|---------------|--------|---------|----------|---------|--------------|--------------|
| time          | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
| 5             | 10     | 1       | 0.900    | 0.0949  | 0.7320       | 1.000        |
| 8             | 9      | 1       | 0.800    | 0.1265  | 0.5868       | 1.000        |
| 10            | 8      | 1       | 0.700    | 0.1449  | 0.4665       | 1.000        |
| 12            | 7      | 1       | 0.600    | 0.1549  | 0.3617       | 0.995        |
| 14            | 6      | 1       | 0.500    | 0.1581  | 0.2690       | 0.929        |
| 16            | 5      | 1       | 0.400    | 0.1549  | 0.1872       | 0.855        |
| 20            | 3      | 1       | 0.267    | 0.1501  | 0.0885       | 0.803        |

| group=Treatment |        |         |          |         |              |              |
|-----------------|--------|---------|----------|---------|--------------|--------------|
| time            | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
| 6               | 10     | 1       | 0.90     | 0.0949  | 0.7320       | 1.000        |
| 12              | 9      | 1       | 0.80     | 0.1265  | 0.5868       | 1.000        |
| 15              | 8      | 1       | 0.70     | 0.1449  | 0.4665       | 1.000        |
| 18              | 7      | 1       | 0.60     | 0.1549  | 0.3617       | 0.995        |
| 28              | 4      | 1       | 0.45     | 0.1743  | 0.2106       | 0.961        |
| 30              | 3      | 1       | 0.30     | 0.1688  | 0.0996       | 0.904        |

```
dat_split <- split(dat, dat$group)
```

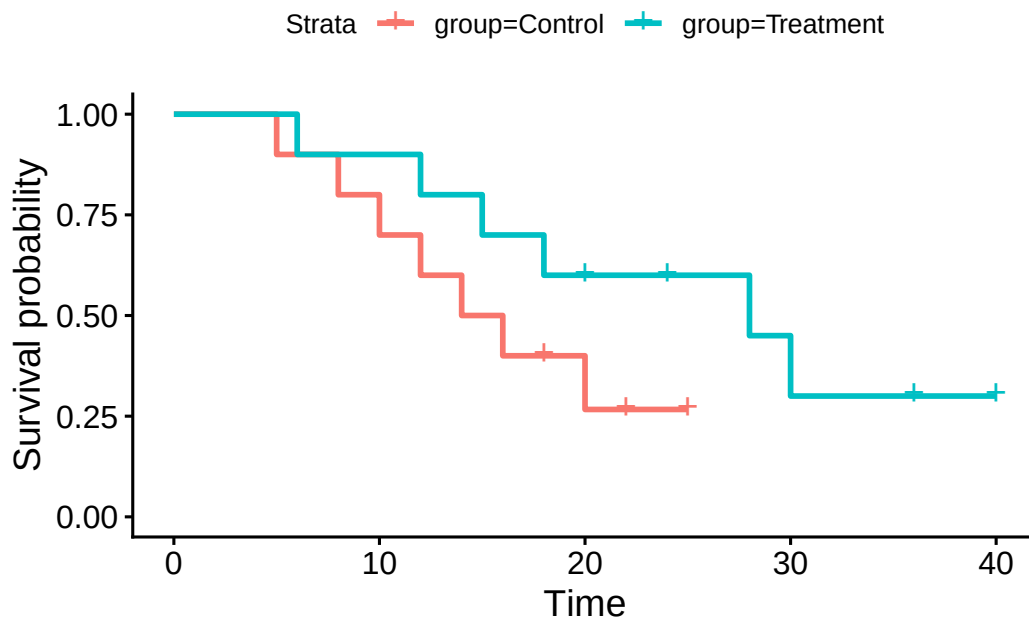
Adding a covariate to `survfit()` just splits the data in two to estimate the survival curves for both populations using the respective samples.

```
summary(survfit(Surv(time, status) ~ 1, data = dat_split[[1]]))
```

Call: `survfit(formula = Surv(time, status) ~ 1, data = dat_split[[1]])`

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 5    | 10     | 1       | 0.900    | 0.0949  | 0.7320       | 1.000        |
| 8    | 9      | 1       | 0.800    | 0.1265  | 0.5868       | 1.000        |
| 10   | 8      | 1       | 0.700    | 0.1449  | 0.4665       | 1.000        |
| 12   | 7      | 1       | 0.600    | 0.1549  | 0.3617       | 0.995        |
| 14   | 6      | 1       | 0.500    | 0.1581  | 0.2690       | 0.929        |
| 16   | 5      | 1       | 0.400    | 0.1549  | 0.1872       | 0.855        |
| 20   | 3      | 1       | 0.267    | 0.1501  | 0.0885       | 0.803        |

```
ggsurvplot(km_fit)
```



```
# Compare survival in both groups
surv_diff <- survdiff(Surv(time, status) ~ group, data = dat)
```

```
surv_diff$var
```

```
      [,1]      [,2]  
[1,]  2.636085 -2.636085  
[2,] -2.636085  2.636085
```

```
# For each time interval (month), get the number at risk, the number of events,  
# and the number of censored observations  
observation_months <- seq(OL, max(dat$time))  
  
r_j <- rep(NA, max(observation_months))  
d_j <- rep(NA, max(observation_months))  
c_j <- rep(NA, max(observation_months))  
  
r_j_trt <- rep(NA, max(observation_months))  
d_j_trt <- rep(NA, max(observation_months))  
c_j_trt <- rep(NA, max(observation_months))  
  
r_j_ctl <- rep(NA, max(observation_months))  
d_j_ctl <- rep(NA, max(observation_months))  
c_j_ctl <- rep(NA, max(observation_months))  
  
for (t in observation_months) {  
  
  r_j[t + 1] <- sum(dat$time >= t)  
  d_j[t + 1] <- sum(dat$time == t & dat$status == 1)  
  c_j[t + 1] <- sum(dat$time == t & dat$status == 0)  
  
  r_j_trt[t + 1] <- sum(dat$time >= t & dat$group == "Treatment")  
  d_j_trt[t + 1] <- sum(dat$time == t & dat$group == "Treatment" & dat$status == 1)  
  c_j_trt[t + 1] <- sum(dat$time == t & dat$group == "Treatment" & dat$status == 0)  
  
  r_j_ctl[t + 1] <- sum(dat$time >= t & dat$group == "Control")  
  d_j_ctl[t + 1] <- sum(dat$time == t & dat$group == "Control" & dat$status == 1)  
  c_j_ctl[t + 1] <- sum(dat$time == t & dat$group == "Control" & dat$status == 0)  
  
}  
  
km_estimates <- tibble(  
  time = observation_months,  
  r_j = r_j,
```

```

d_j = d_j,
c_j = c_j,
r_j_trt = r_j_trt,
d_j_trt = d_j_trt,
c_j_trt = c_j_trt,
r_j_ctl = r_j_ctl,
d_j_ctl = d_j_ctl,
c_j_ctl = c_j_ctl
)

# Calculate the expected number of events for each group at each time point
km_estimates <- km_estimates |>
  mutate(
    d_j_trt_exp = r_j_trt * (d_j / r_j),
    v_j_trt = (r_j_trt * r_j_ctl * d_j * (r_j - d_j)) / (r_j^2 * (r_j - 1)),
    d_j_ctl_exp = r_j_ctl * (d_j / r_j),
    v_j_ctl = (r_j_trt * r_j_ctl * d_j * (r_j - d_j)) / (r_j^2 * (r_j - 1))
  ) |>
  filter(d_j > 0)

```

```
sum(km_estimates$d_j_ctl)
```

```
[1] 7
```

```
sum(km_estimates$d_j_ctl_exp)
```

```
[1] 4.746096
```

```
(sum(km_estimates$d_j_ctl) - sum(km_estimates$d_j_ctl_exp))^2 / sum(km_estimates$v_j_ctl, na.rm=T)
```

```
[1] 1.927133
```

```
sum(km_estimates$d_j_trt)
```

```
[1] 6
```

```
sum(km_estimates$d_j_trt_exp)
```

```
[1] 8.253904
```

```
(sum(km_estimates$d_j_trt - km_estimates$d_j_trt_exp))^2 / sum(km_estimates$v_j_trt, na.rm =
```

```
[1] 1.927133
```